

Shared Hub: Training, Transfer and Retrieval

A consolidated summary of this conversation

Prepared for Liad Kashanovsky | 15 September 2026

One shared coordinate system; one entry map per input encoder; a separately trained caption head. This guide explains the full training and prediction process, using paired cat and dog image-caption items.

The essential pipeline



The entry map outputs the representation that the caption head receives.



Output is a vector used to retrieve a stored caption, not generated text.

Image → encoder embedding → entry map → predicted hub vector → caption head → predicted bge vector → ranked captions

Main distinction: reference hub coordinates H supervise entry-map training. DINO-small's predicted coordinates $H_{\text{small}} = X_{\text{small}} W_{\text{small}}$ supply the inputs for caption-head training.

Scope and evidence. This is a corrected synthesis, not a verbatim transcript. Reported scores come from the supplied Hub_Scores_Summary.pdf, with more precise diagnostic values and setup details from the visible conversation. The underlying notebook and full report were not independently inspected. Illustrative cluster figures are synthetic, not experimental projections.

Reading order: terminology → encoder inventory → construction and training → caption head → prediction and ConvNeXt transfer → readability/writability → scores → corrections and interpretation.

1. Terms, references and locations

Term / symbol	Meaning	Cat example
Item i	One paired dataset row	One specific cat image and its matching caption.
Encoder	Pretrained model that produces an embedding	DINO-small for the image; bge for its caption.
Representation / embedding	A vector of numbers	x_{small} has 768 components in this experiment.
Coordinates / location	Numbers / the point those numbers specify in a space	512 hub coordinates specify one point.
Reference h_i	The hub-construction output for item i	The target hub location for this particular cat item.
Predicted $\hat{h}_{k,i}$	Encoder k 's estimated hub location	$x_{\text{small},i} W_{\text{small}}$; usually not exactly h_i .
Actual t_i	Caption embedding produced by the target encoder	$\text{bge}(\text{"a cat sitting on a sofa"})$.
Predicted $\hat{t}_i$	Caption-head output	A vector used to rank stored captions.
X, H, T	Matrices: one item per row	Many image vectors, hub locations or caption vectors.
W_k / W_{head}	Entry / output matrices	DINO-small $\rightarrow$ hub / hub $\rightarrow$ bge.

Reference is not universal ground truth. It is the location defined by the selected fitted hub. There is no manually designated cat section, GPT-2 section, or DINO-small section inside that hub.

One item, several representations. The same cat item has a DINO-small embedding, a bge caption embedding, a reference hub vector, and one predicted hub vector per entry route. Related cat items may form a cluster, but each item has its own location.

Route versus destination: the encoder determines which W_k to apply; the vector's content determines the predicted location. W is shared across all items, not one matrix per cat, dog, or category.

2. Encoders and hub configurations

The supplied summary distinguishes a published four-space hub from a seven-space variant. A frozen hub means one chosen configuration remains fixed while new routes or queries are evaluated.

Encoder	Input	Width	4-space	7-space	Entry W shape
DINOv2-small	Image	768	Yes	Yes	768×512
DINOv2-base	Image	1536	Yes	Yes	1536×512
DINOv2-large	Image	2048	Yes	Yes	2048×512
bge-m3	Text	1024	Yes	Yes	1024×512
SBERT	Text	768	No	Yes	768×512
BERT	Text	768	No	Yes	768×512
GPT-2	Text	768	No	Yes	768×512
ConvNeXt-base	Image	1024*	No	No	$1024 \times 512^*$

Widths are stored feature widths reported for this experiment, not universal specifications for every model configuration. *ConvNeXt's 1024 width comes from the described setup in the conversation.

Four-space concatenation: $768 + 1536 + 2048 + 1024 = 5376$

Seven-space concatenation: $5376 + 768 + 768 + 768 = 7680$

Both are projected into a 512-dimensional hub.

Correction: earlier wording assigned 5376 dimensions to the seven-space concatenation. That is the four-space total; the seven-space total is 7680.

What does W mean?

$$\mathbf{x}_{\text{small}} [1 \times 768] \cdot \mathbf{W}_{\text{small}} [768 \times 512] = \mathbf{h}_{\text{small}} [1 \times 512]$$

W is a matrix of learned coefficients. Each hub coordinate is a weighted combination of the input coordinates. Fitting W does not retrain the original encoder. The output matrix may be called V_bge or W_head: the names differ, but its input and output define its role.

An output map is not automatically the inverse of an entry map. The matrices are separately fitted, can be rectangular, and operate under information loss and regularization.

3. Build the hub and fit entry maps

Previously stated data split: 9533 paired items, with 8533 used for fitting and 1000 held out for evaluation. The two-page score summary does not itself verify every split and preprocessing detail.

Stage	Input	Fitted component / target	Output
Extract	Training images and matching captions	Frozen pretrained encoders; no encoder fine-tuning	One X_k per encoder
Standardize	Member features X_k	Training-set means and scales for each encoder	$Z_k(X_k)$
Concatenate	Standardized member vectors for the same item	No separate predictive fit	C: 5376 or 7680 columns
PCA + whiten	C on fitting items	PCA basis and whitening scales	Reference H: 8533×512
Fit entry k	Encoder k features on fitting items	Ridge W_k ; target is the corresponding H rows	$W_k: d_k \times 512$
Apply entry	Features X_k	No new fit	$H_k = X_k W_k$

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C = concat(Z_1(X_1), ..., Z_K(X_K))
H = Whiten(PCA_512(C))
W_k = argmin_W [ ||X_k W - H||^2_F + \alpha_k ||W||^2_F ]
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X_k in the map equations denotes the actual features supplied to the fitted entry pipeline, including any required preprocessing. The conversation does not establish every intercept or normalization detail; those must match the notebook.

For standard ridge without an intercept, the solution can be expressed as $W_k = (X_k^T X_k + \alpha_k I)^{-1} X_k^T H$. Implementations normally solve the linear system rather than explicitly forming an inverse.

What is learned from a cat and a dog?

A paired cat row supplies $(x_{k,cat}, h_{cat})$; a dog row supplies $(x_{k,dog}, h_{dog})$. One matrix W_k is fitted across all training items so its outputs approximate the corresponding reference vectors. Class labels are not required by this paired regression.

Read direction: encoder $\rightarrow$ hub. Readability asks whether this route recovers useful item information. Constructing H is a prior, separate step; H is not built by averaging the later entry-map predictions.

Evaluation discipline: fit standardization, PCA, whitening and regressions on fitting data, then freeze them. Test targets may be used to score performance, not to refit the pipeline.

4. Train the caption output head

After the reference hub and DINO-small entry map are fitted, obtain the representations that the caption head will actually receive:

$$H_{\text{small}} = X_{\text{small}} W_{\text{small}}$$

$$W_{\text{head}} = \operatorname{argmin}_W [\|H_{\text{small}} W - T_{\text{bge}}\|_F^2 + \alpha_{\text{head}} \|W\|_F^2]$$

Component	Role in head training	Shape for 8533 fitting rows
X_{small}	DINO-small image features	8533×768
W_{small}	Already-fitted entry map	768×512
H_{small}	Predicted hub coordinates; training INPUT	8533×512
T_{bge}	Actual matching-caption embeddings; training TARGET	8533×1024
W_{head}	Learned output matrix	512×1024
$H_{\text{small}} W_{\text{head}}$	Predicted caption embeddings	8533×1024

Why DINO-small? It is the chosen training-source route in this experiment, not an intrinsic requirement of the hub. Choosing one source makes it possible to test whether the same frozen head works through other encoders. Other source choices require separate experiments.

Why not feed H directly? This setup trains on H_{small} so the head sees the representations produced by the DINO-small route, including its approximation errors. Training directly on reference H would be a different design.

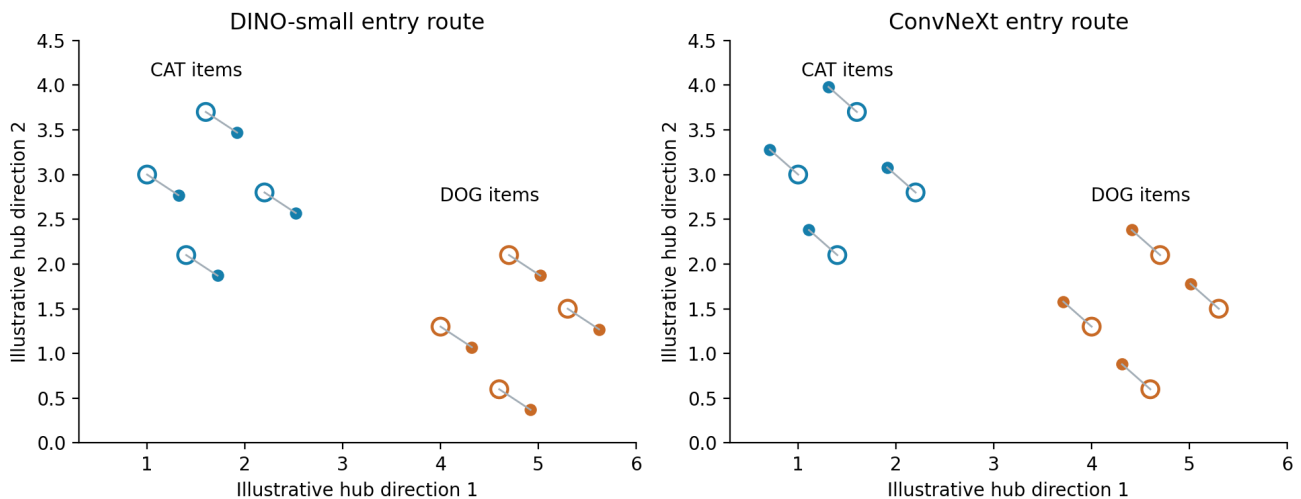
Why bge? bge supplies the target caption embedding space. The head learns image-derived hub coordinates $\rightarrow$ bge vectors. DINO-small and bge are not merged into a new encoder; they define the input route and the target supervision, respectively.



Output is a vector used to retrieve a stored caption, not generated text.

Writing is not text generation. After predicting a bge vector, the system ranks stored captions by cosine similarity. Generating a new sentence would require an additional decoding mechanism.

5. Cat-dog clusters: same item, two locations



Legend: hollow circle = reference $h_{i,j}$; filled point = predicted $\hat{h}_{k,i}$; connecting line = entry-map error for the same item. Colors distinguish cat and dog items. These are synthetic illustrations, not actual model outputs, errors, or measured clusters.

Both panels use the same reference locations. Only the predicted locations differ between entry routes. The real hub has 512 coordinates; this diagram merely illustrates the relationship in two dimensions.

What the caption head sees

During its original training, the head receives the filled DINO-small points, paired with actual bge caption vectors. During ConvNeXt transfer, it receives the filled ConvNeXt points while its coefficients remain unchanged.

What cluster proximity does and does not establish

A cat prediction near a cat cluster illustrates semantic plausibility. Retrieval R@1 is stricter: the system must rank the specific matching caption above all other candidates, potentially including many other cats. Good class clustering alone does not establish high instance-level retrieval.

A predicted hub vector is a numerical representation, not a route selector. The encoder chooses the entry matrix; the input content determines where that matrix places the item.

6. Use the frozen system and add ConvNeXt

Step	Original member: DINO-small	Held-out encoder: ConvNeXt
Before querying	Existing entry map is already fitted	Fit ONE entry map to frozen H using fitting items
Image input	A new cat image	A new dog image
Encode	$x_{\text{small}}: 1 \times 768$	$x_{\text{conv}}: 1 \times 1024$
Enter hub	$x_{\text{small}} W_{\text{small}} \rightarrow 1 \times 512$	$x_{\text{conv}} W_{\text{conv}} \rightarrow 1 \times 512$
Write into bge	Apply frozen $W_{\text{head}} \rightarrow 1 \times 1024$	Apply SAME frozen $W_{\text{head}} \rightarrow 1 \times 1024$
Retrieve	Cosine against stored bge captions; rank	Same candidate-space procedure
Refit per query?	No	No

Member route: $\hat{t} = x_{\text{small}} W_{\text{small}} W_{\text{head}}$

Held-out route: $\hat{t} = x_{\text{conv}} W_{\text{conv}} W_{\text{head}}$

$\text{score}(j) = \text{cosine}(\hat{t}, t_j)$; return the highest-ranked candidate(s).

DINO-base and DINO-large replace W_{small} with their own entry matrices and reuse the head. A text encoder also has an entry route, but applying an image-trained head to text-derived hub vectors is a transfer condition to evaluate, not a guaranteed success.

Two meanings of held out

Encoder held out: ConvNeXt did not participate in hub construction or original caption-head training.

Items held out: test image-caption pairs must not be used to fit its new entry map.

Zero-shot head transfer: the caption head sees no ConvNeXt examples during its training. This is not zero-data adaptation: the ConvNeXt entry map is fitted on paired training items.

The hub is not rebuilt. The caption head is not retrained for the transfer route. The new input does not require all seven encoders or a freshly constructed reference H. Existing entry maps and target caption embeddings are reused.

Additional fits used only for evaluation

Native/reference heads are fitted specifically for each encoder to form comparison denominators. Separate output maps are also needed when evaluating SBERT, BERT or GPT-2 as targets. These comparison fits are not necessary to run the single frozen bge-head transfer pipeline. Their exact protocols require the full report or notebook.

7. Readable, writable and anisotropy

Question	Readable	Writable
Direction	Encoder space → hub	Hub → target space
Map	Entry W_k	Output $W_{out,target}$
Input	Actual encoder embedding	Hub representation
Output	Predicted hub coordinates	Predicted target embedding
Success requires	Recoverable item information	Prediction precise enough for correct ranking
Cat/dog example	Extract differences in actual vectors	Predicted cat vector must outrank dog and other captions

GPT-2's actual vectors can preserve small informative differences even if most point in nearly the same direction. A learned map can exploit those differences when reading. Predicting back into a tightly aligned space can be sensitive to error: a slight shift may reverse the candidate ranking.

Space	Read score	Write / target score	Pair cosine
bge-m3	98.1%*	98.1%	+0.570
SBERT	Not supplied	101.0%	+0.135
BERT	Not supplied	76.6%	Not supplied
GPT-2	95.1%*	24.8%*	About +0.999*

*Precise values from the conversation; the PDF rounds GPT-2 to about 95% readable and 25% writable. These relative scores are not absolute retrieval accuracy. The exact diagnostic normalization and training inputs were not recoverable from the two-page summary.

Effective rank: a continuous measure of how concentrated a spectrum is, not a literal number of semantic dimensions. Reported values in the conversation were GPT-2 6.6 and bge 97.4. Interpretation depends on whether features were centered or otherwise transformed before computing the spectrum.

Average pair cosine: describes an aggregate directional tendency; it is not the measured angle of every cat-dog pair. High cosine does not by itself prove all rankings are impossible. Anisotropy is a plausible contributor to sensitivity, not a demonstrated sole cause of the observed score differences.

A controlled centering/whitening comparison, fitted on training data and evaluated for both transferred and native heads, could test the proposed mechanism. Such a comparison is a suggested follow-up, not a result established here.

8. Reported performance and protocols

Source: supplied Hub_Scores_Summary.pdf. Numbers are reported, not independently rerun. Its table combines distinct metric types; the separation below avoids treating them as interchangeable.

Route	Raw R@1	% of native	Role
DINO-small	0.398	111.5%	Head-training source
DINO-base	0.442	95.9%	Member head transfer
DINO-large	0.468	92.9%	Member head transfer
GPT-2 transfer	0.095	27%	Different from its ~25% writability
Random map	0.000 observed	—	Chance expectation: 0.001

Held-out encoder	Published	Rebuilt	Interpretation
SigLIP 2	94.2%	86.4%	Separate model lineage
ConvNeXt-base	96.7%	87.5%	Convolutional architecture
DINOv2-giant	100.0%	—	Summary marks result inconclusive
Qwen3-Emb-0.6B	93.7%	—	Text-to-text, not image-to-text
Qwen3-Emb-4B	96.2%	—	Summary marks result inconclusive

$R@1 = \text{correct top-ranked queries} / \text{total queries}$

$\text{Retention (\%)} = 100 \times \text{frozen-head } R@1 / \text{native-reference-head } R@1$

A 96.7% retention score is not 96.7% absolute accuracy. The empirical native reference is not a strict ceiling, so values above 100% are possible. Compare within a protocol: published, rebuilt and scale-extension scores differ.

The summary additionally reports ConvNeXt at 101.2% in a scale protocol, with McNemar $p = 0.668$. That result does not establish an improvement over its comparator. Its statement of a protocol offset should not be used to convert arbitrary scores between runs.

Text participation: G10 reports 85.5% of native for an images-only hub and 95.3% after adding bge, a 9.8-point increase in that comparison. This does not establish that adding every text encoder always helps.

9. Corrections, conclusions and sources

Corrections to retain from the discussion

Earlier wording	Correct interpretation
Seven-space width is 5376	5376 is four-space; seven-space is 7680.
Smaller denominator makes retention conservative	At fixed numerator, a smaller denominator raises the ratio. It does not prove conservatism.
Changing target changes only denominator	Both transferred-head and native-head performance can change.
Anisotropy cancels automatically	Cancellation requires equal multiplicative effects; evidence is needed.
SBERT is about 3 points better on both metrics	Writability gap: 2.9 points. Raw R@1 gap 0.510 – 0.466: 4.4 points.
GPT-2 can never be a target	It performs poorly under this reported mapping/ranking setup.
Low effective rank literally means 6.6 content dimensions	Effective rank is a spectral summary, not a literal count.
Three documents already agree	The report, Defense_Brief and Mock_Viva were not inspected here.

Illustration of the ratio correction: a fixed system score of 45% gives 90% retention against a 50% reference, but 75% against a 60% reference. This is arithmetic, not an empirical prediction about switching bge to SBERT.

Other experimental context in the supplied summary

Scale extension reports raw cross-modal shape agreement rising then falling: DINO large $p = 0.418$ versus giant 0.339; contrastive text peak Qwen-0.6B 0.352 versus Qwen-4B 0.307. These are no-map shape-agreement statistics, not retrieval or retention. The summary labels the tested monotonic-scaling predictions falsified within its examined regime; that does not settle larger-model regimes.

Sources and limits

1. Hub_Scores_Summary.pdf, supplied attachment, 2 pages: architecture, reported transfer scores, protocol distinctions, diagnostics and scale-extension summary.
2. This conversation: split, equations, map dimensions, detailed terminology, precise read/write values and corrections.
3. Ethayarajh (2019), How Contextual are Contextualized Word Representations? General anisotropy background: aclanthology.org/D19-1006. It does not validate this project’s particular scores or causal claims.

Final formulation: H defines the reference coordinate system. $X_k W_k$ predicts a location in it. $X_k W_k W_{head}$ predicts a bge caption embedding. Retrieval ranks actual captions against that prediction. The entry route changes with the input encoder; the frozen head can be reused and its transfer performance measured.